

Ahmed Abdelrahman

Cairo, Egypt (UTC+3) | +201024180920 | ahmedabdelrhman232@gmail.com |
https://linkedin.com/in/ahmed-abdelrahman | https://kennycandra.dev

Skills

languages

TypeScript, JavaScript, Rust, Python

frontend

React, Next.js, Tailwind CSS, Redux

backend

Node.js, Express.js, REST APIs, Event-Driven Architecture, Async Processing, Software Design

databases

PostgreSQL (pgvector), MongoDB, MySQL, Redis (Caching & Queues)

asyncAndMessaging

BullMQ, RabbitMQ, Dead-Letter Queues, Idempotent Processing, Exponential Backoff

infrastructure

AWS (ECS Fargate, ECR, RDS, ElastiCache, Lambda, SSM), Docker, GitHub Actions CI/CD, Vercel, DigitalOcean

coreCompetencies

Distributed Systems, System Design, CAP Theorem, MVCC, Database Internals, Networking (HTTP/2, HTTP/3, TCP/UDP, WebSockets)

spoken

English (Professional), Arabic (Native)

Education

Faculty of Pharmacy

2024 Modern University, Cairo, Egypt

Work Experience

Vogel Edits

Full-Stack Engineer (Adobe Extensions)

- Delivered customer-facing AI extensions for Adobe After Effects & Premiere Pro, including Vanish BG — a real-time background removal tool — serving as the technical bridge between ML capabilities and end-user production workflows.
- Debugged and resolved Adobe's restricted CEP runtime limitations by designing a Node.js sidecar architecture that handles AI inference outside the sandbox, using local HTTP for IPC with real-time streaming.
- Built and shipped a TTS Premiere Pro extension using Node.js with local ONNX model inference, owning the full pipeline from text input to rendered timeline clip.
- Collaborated directly with designers and stakeholders in an async workflow, triaging customer issues and shipping production-ready tools on agile cycles.

Freelance

Full-Stack Engineer

- Built and integrated a full-stack tourism booking platform using Next.js, TypeScript, and Node.js — handling end-to-end implementation from customer-facing UI to backend APIs.
- Diagnosed and resolved production performance bottlenecks using PostgreSQL EXPLAIN ANALYZE, achieving sub-100ms API response times under concurrent load.
- Integrated Paymob payment gateway with idempotent transaction handling and webhook verification, reducing checkout friction by 40% — directly analogous to enterprise third-party integrations.
- Optimized Core Web Vitals to 95+ Lighthouse score, improving customer experience through SSR, image optimization, and lazy loading.

Projects

RootCluster — AI-Powered Log Analysis Platform <https://rootcluster.dev>

Tech Stack: Node.js, TypeScript, PostgreSQL (pgvector), Redis, BullMQ, ONNX, AWS ECS Fargate, Docker, GitHub Actions

- Designed and deployed distributed log ingestion pipeline on AWS ECS Fargate processing 1M+ events/hour with sub-second latency.
- Reduced ONNX embedding time from 20+ minutes to under 1 minute via batch processing, Redis caching, and worker autoscaling.
- Designed event-driven pipeline: normalize → embed → cluster → merge → root cause, with retries, exponential backoff, DLQs, and idempotent processing.
- Implemented centroid-based pgvector clustering to unify cross-service failures into single root causes.
- Set up full CI/CD with GitHub Actions OIDC, ECR, SSM Parameter Store, RDS PostgreSQL, and ElastiCache.
- Applied PostgreSQL internals (MVCC, isolation levels, indexing) to build high-throughput write pipeline avoiding lock contention.
- Leveraged CAP theorem tradeoffs: PostgreSQL for strong consistency, Redis for high-availability caching.

Medical CRM — AI-Enhanced Prescription Platform

Tech Stack: React, Node.js, PostgreSQL

- Built full-stack CRM handling diagnoses, prescriptions, and patient records for healthcare providers.
- Implemented JWT auth with access/refresh token rotation and RBAC across three permission levels.
- Designed drug interaction validation across 500+ medications using graph-based PostgreSQL lookup.
- Optimized dashboard queries for 10K+ monthly records using composite indexes and window functions.